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Particle impact ignition in high-pressure oxygen: Experiments, mechanisms, and multiphysics modeling

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ABSTRACT

Particle impact ignition in high-pressure oxygen poses a significant threat to oxygen-rich turbomachinery, yet current experimental tests cannot access the temperatures and pressures encountered in service, limiting their predictive value. This study addresses this challenge through a combined experimental-numerical approach in which particle impact tests are used to calibrate a multiphysics model that predicts ignition under engine-relevant conditions beyond those explicitly tested. Experiments with 100-μm Ti-6Al-4V particles impacting Al_2O_3 and Ni targets reveal a critical impact velocity for ignition that decreases with increasing gas temperature and target hardness. At 300 K, no particle ignitions are observed on Ni, while the critical velocity on Al_2O_3 is 170 m/s; at 500 K, the respective critical velocities are 225 m/s and 84 m/s. The model, anchored by these data, incorporates plasticity, adiabatic heating, and oxidation-driven thermal effects to simulate impact and ignition behaviors. A parametric study over ranges of temperatures, pressures, and particle sizes relevant to rocket engine applications shows that increasing these parameters lowers the critical velocity by enhancing plasticity, oxide rupture, and localized heat generation, with especially high ignition risk above 600 K and 6 MPa, conditions common in oxygen-rich turbomachinery. These findings clarify the governing mechanisms of particle impact ignition, highlight the relative benefits of different mitigations such as soft inert coatings, and establish a predictive framework for ignition risk assessment in high-pressure oxygen systems.

Novelty and significance statement

This work develops a quantitative link between impact and ignition of metal particles in high-pressure oxygen. In a novel approach, numerical modeling integrates impact physics, such as plasticity and adiabatic heating, with a temperature-dependent metal oxidation model to reproduce impact-driven ignition. The numerical model is calibrated with experiments and can be used to predict the risk of particle ignition under elevated temperatures and pressures relevant to rocket engine operating conditions which are inaccessible with current test methods.

1. Introduction

In the turbine of an oxygen-rich turbopump, temperatures and pressures are typically in the range 450–900 K and 15–60 MPa [1], and in these extreme operating conditions, metal hardware is susceptible to ignition [2,3]. In this work, we focus on particle impact ignition, represented in Fig. 1 [4], where metal particles entrained in the gas flow strike a component at high velocities, ignite, and then kindle to downstream hardware, causing catastrophic engine fires [5]. To avoid this failure mode, rocket engine design currently relies on qualitative heuristics and extensive testing, which gives limited insight as tests cannot currently access the temperatures and pressures expected in service. These limitations are especially critical given the recent push for reusable launch vehicles, which accumulate particulate and debris over multiple flight cycles and incorporate additively manufactured

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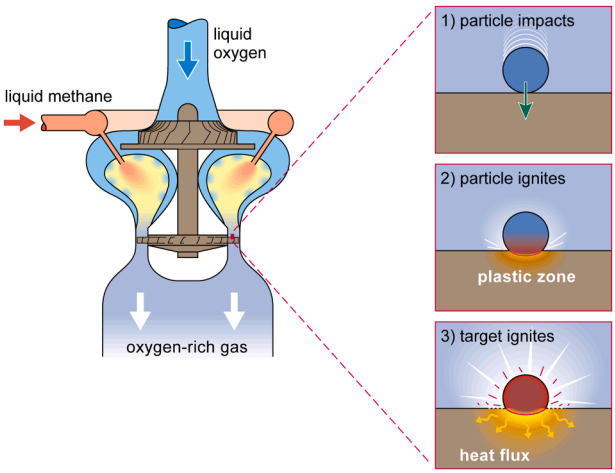

Fig. 1. Particles within an oxygen-rich turbopump can strike component surfaces at high velocities, causing the particle to ignite and kindle to the component.

hardware that can shed powder particles [6].

The risk of particle impact ignition has historically been assessed using particle impact experiments in high-pressure oxygen, which accelerate particles through various mechanisms including supersonic gas flow [4,7], subsonic gas flow [8], and propagating shock waves [9]. Typical particle materials include reactive metals (e.g., Al, stainless steel, or Ti alloys) [4,7,10–17], with particle sizes in the range 100–2000 μm, while target materials include common turbomachinery alloys (e.g., Inconel alloys [4,7,12,13,18,19], Monel alloys [7,10–13], stainless steels [4,7,10–12,14,17], etc.). Target samples are analyzed post-mortem, and results are grouped qualitatively based on the severity of ignition. Despite the large volume of test data, there has been little effort to quantitatively relate particle impact to the observed ignition outcomes. Additionally, test pressures and temperatures are typically below the turbine inlet conditions in modern rocket engines, as shown in Fig. 2. As a result, findings from particle impact experiments do not relate directly to in-service ignition risk.

These particle impact experiments have nevertheless revealed the main processes comprising particle impact ignition, illustrated in Fig. 1, as well as key trends in target ignition. For example, Fe-based and Al-based alloys ignite readily at temperatures and oxygen pressures above 400 K and 3.5 MPa [4,10,11,18], whereas Cu-based alloys are

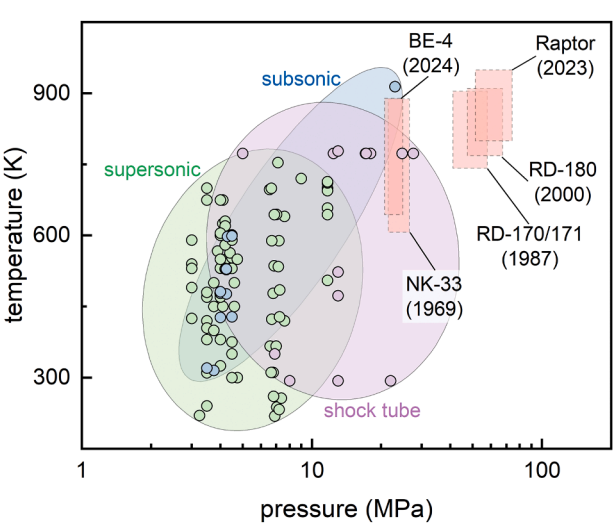

Fig. 2. Accessible temperatures and pressures for subsonic, supersonic, and shock tube tests are limited relative to turbine operating conditions [4,7–9].

more ignition-resistant, with few ignition events up to 550 K and 30 MPa [7,20]. The particle material has also been found to strongly influence target ignition outcomes: Al particles, which exhibit a characteristically high energy release rate during burning, promote target ignition more readily than SS316 and IN718 particles [4,21]. These findings link ignition outcomes to material properties: ignition risk increases with the particle and target enthalpy of oxidation and decreases with the target thermal effusivity, which each influence local heating and the rate of heat accumulation [4,12]. Several studies have drawn on these findings to develop empirical material rankings [4,7,18,22], but the quantitative relationship between material behavior and ignition outcomes has not been explained. And while these studies emphasize behavior of the target, they often disregard particle ignition, which is a critical first step in the kindling chain that eventually leads to target ignition.

The central hypothesis of the present work is that particle ignition is driven by high-rate plasticity and adiabatic heating upon impact. Previous studies have developed frameworks to describe particle ignition under static conditions, and they find that it occurs when the rate of heat accumulation is net positive. For metallic systems, heat accumulation is governed by a competition between exothermic oxidation and combined heat losses from convection, conduction, and radiation, which each depend on geometry, ambient temperature and pressure, and gas flow [23,24]. In static conditions, without considering deformation processes, the rate of metal oxidation is limited by the presence of a surface oxide, so ignition in oxygen-rich environments occurs when the oxide is disrupted [25]. Oxides can be disrupted by phase transformations (e.g., metal or oxide melting or solid-state phase transformations) or through mechanical rupture (e.g., due to thermal expansion mismatch between the oxide and metal substrate) [26,27]. Under impact conditions, we expect that ignition results mainly from mechanical rupture of the native oxide, similar to other mechanically driven ignition phenomena, such as fracture or frictional ignition [28,29]. Additionally, mechanical rupture of the oxide overlayer exposes bare metal to the oxygen-rich environment, where the oxidation rate is controlled by oxygen adsorption on the surface [30]. As a result, the activation energy of mechanically driven oxidation is roughly an order of magnitude less than that reported for static oxidation [30,31], exacerbating the risk of ignition.

In this work, we consider these insights together with experiments to develop a numerical model that can assess particle ignition risk under conditions directly relevant to those of oxygen-rich turbomachinery. We perform particle impact experiments in high-pressure oxygen to determine critical ignition velocities for Ti-6Al-4V particles under a range of temperatures for two different target materials. We then use the experimental results to calibrate a multiphysics numerical model that describes particle ignition upon impact, and we extend the model over a range of temperatures, pressures, and particle sizes relevant to application. This work focuses on particle ignition, a critical first step in the process of particle impact ignition (cf. Fig. 1); future work will model the kindling of burning particles to a target. These modeling approaches can be integrated to predict component-level ignition risk in oxygen-rich turbomachinery.

2. Particle ignition experiments

We performed particle impact experiments in flowing oxygen to determine the effects of temperature and target material on the critical velocity for particle ignition. These experiments were performed at the NASA White Sands Test Facility with the subsonic particle impact setup shown in Fig. 3. In these tests, oxygen is preheated and directed into the gas inlet at subsonic velocities relative to the speed of sound in oxygen gas at the test conditions. Hemispherical cups separate to release 5 mg of particles (around 2000 particles) into the flowing gas. The gas then travels through the nozzle and accelerates the particles into the target at velocities up to 400 m/s. To capture impact velocity, rebound velocity, and particle ignition outcomes (i.e., ignition or no ignition), we recorded videos using a high-speed camera with an interframe time of 34 μs and

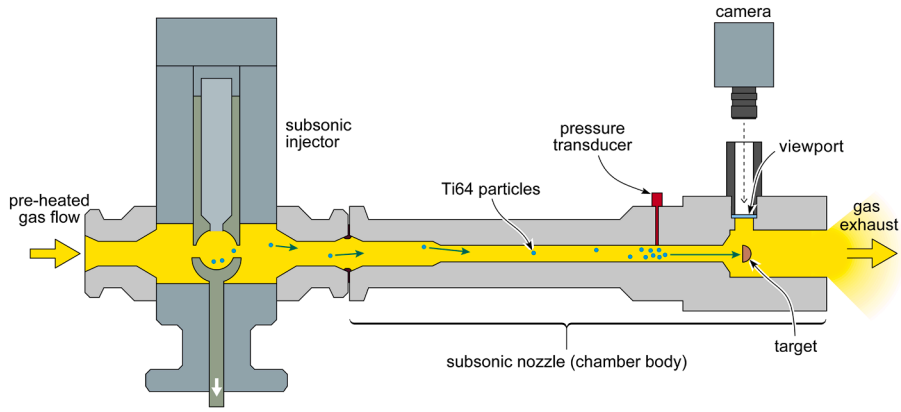

Fig. 3. Particle impact experiments in high-pressure, flowing oxygen capture particle ignition under a range of impact velocities and temperatures.

exposure time of 0.97 μs, aimed through a viewport orthogonal to the particle trajectory. The tests were illuminated using a light aimed through the same viewport as the high-speed camera. For these tests, we used Ti-6Al-4V, wt % (Ti64) particles, which are ignition prone due to their high enthalpy of oxidation and low thermal effusivity (1.97×10^7 J/kg and 4200 $J/m^2 \cdot K \cdot s^{1/2}$, respectively) [32,33]. The particles were sieved to sizes between 90 and 106 μm with a mean size of 100 μm. We used two different target materials – Ni and Al_2O_3. Through nano-indentation testing, we measured hardness values of 2.6 GPa and 28.1 GPa for the Ni and Al_2O_3 targets and 5.7 GPa for the Ti64 particles. This range in target hardness means that the present material combinations give limiting physical bounds, where Ti64 impact on Ni causes extensive plastic deformation in the target while impact on Al_2O_3 isolates plastic deformation in the particle.

For both target materials, we performed sets of low-temperature (~300 K) and high-temperature (~500 K) tests, repeating each test condition >25 times. Table 1 gives the ranges of test conditions over all the tests. Gas temperature was measured before each test, and the particles and target were preheated by the flowing gas for several minutes. Accordingly, we assumed that their temperatures were equal to the gas stagnation temperature. We varied gas velocity between tests by adjusting the width of a downstream orifice, estimating the average gas velocity for each test using an analytical model calibrated to the test rig with CFD, which did not consider temperature effects and may under-predict gas velocity at high temperature. The average gas velocities ranged from 45 to 285 m/s, varying up to ±33 % radially through the test chamber, and we estimated the resulting stagnation pressures P_0 at the target face using

$$P_0 = P + 1/2\rho v^2, \tag{1}$$

with static pressure P, measured upstream of the target, oxygen density ρ [34], and nominal gas velocity v. The values reported in Table 1 correspond to gas temperatures measured directly before testing and stagnation pressures estimated using Eq. (1). Beyond these reported values, CFD predicts that pressure and temperature can vary over the target surface by ±33 % and ±25 % respectively.

Table 1
Low-temperature and high-temperature particle impact test conditions for Ti64 particles on Al_2O_3 and Ni targets.

	Al_2O_3 target		Ni target	
	low temp.	high temp.	low temp.	high temp.
number of tests	43	39	31	31
temperature range (K)	293–304	469–505	277–291	477–485
mean temperature (K)	298 ± 3	481 ± 5	285 ± 5	479 ± 2
pressure range (MPa)	2.1–2.7	1.7–2.9	2.1–2.6	2.1–2.7
mean pressure (MPa)	2.4 ± 0.2	2.2 ± 0.2	2.3 ± 0.2	2.4 ± 0.2

In-situ images taken from a video of a high-temperature test on an Al_2O_3 target are shown in Fig. 4. Here, a moving particle is highlighted by red crosshairs, with the direction of particle motion denoted by a white arrow. The particle impacts the target surface, outlined by the dashed white line, and rebounds. Note that features of the test rig are visible in the background but have no effect on the test. To determine particle velocities from the test videos, we measured the distance traveled by particles between frames over the interframe time of 34 μs. Fig. 4a shows a case where the particle does not ignite: the particle impacts at 69 m/s and rebounds from the target at 16 m/s. Fig. 4b shows a case of particle ignition: the particle impacts at 228 m/s and emits light that intensifies as the particle ignites and burns.

When ignition occurs, particles strike the target and ignite within a single frame, which indicates that the ignition delay time is <34 μs. Additionally, some particles briefly emit light upon impact without fully igniting, which is likely a sign of jetted material. Several particles in each test ignite before entering the frame. Impacts are generally perpendicular to the target. A small fraction of particles impacted at oblique angles, greater than ~10° to the target normal. The test videos do not capture all impacts and ignition events as we observe hundreds of particles per test travel out of focus or out of frame during flight. Notably, for the low-temperature tests on Ni, several ignition events occurred out of frame while none were captured in frame. From the test videos, we measured particle impacts and ignition outcomes for around 100 impacts per test condition.

The aggregated ignition outcomes are shown in Fig. 5, which gives coefficient of restitution (COR) vs. impact velocity, where COR is the ratio of rebound velocity to impact velocity. Particle ignition events are denoted in red and are plotted directly on the x-axis because we could not measure a rebound velocity when particles ignited. We find that COR decreases with impact velocity, which aligns with analytical predictions and experiments that show the relationship COR $\propto v_i^{-1/2}$ [35]. For each test condition, we defined the minimum impact velocity that caused ignition as a critical velocity v_{crit}, indicated by a vertical red line.

Fig. 5 shows that v_{crit} is generally lower in high-temperature tests and on the harder Al_2O_3 target (170 m/s at 300 K and 84 m/s at 500 K), which highlights a connection between particle deformation upon impact, adiabatic heating, and ignition. Along with these results, Fig. 5 shows COR measurements and critical velocities predicted by numerical modeling, discussed in Section 3.

From these results, we see that the ignition outcomes within each test condition vary significantly. This variability arises in part from the slight differences in temperature and pressure between tests, as summarized in Table 1. But more importantly, there is significant uncertainty within each test that is not captured by the temperature and pressure measurements. For example, CFD predicts that high-speed gas flow around

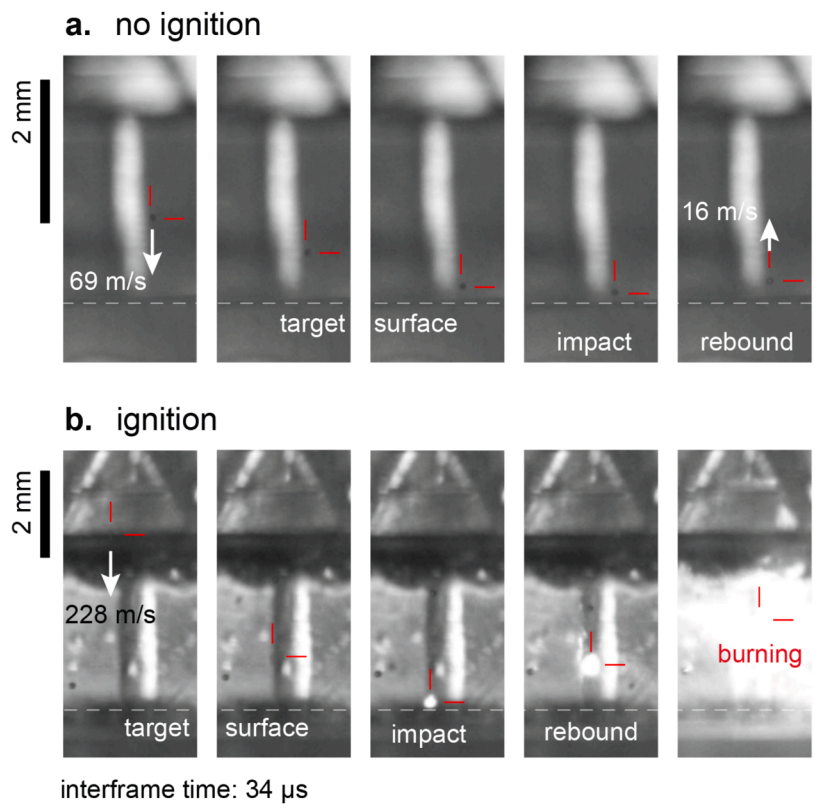

Fig. 4. Examples of Ti64 particle impact on Al_2O_3 at high temperature showing a) no ignition, where the particle impacts at 69 m/s and rebounds at 16 m/s, and b) ignition, where the particle impacts at 228 m/s and ignites.

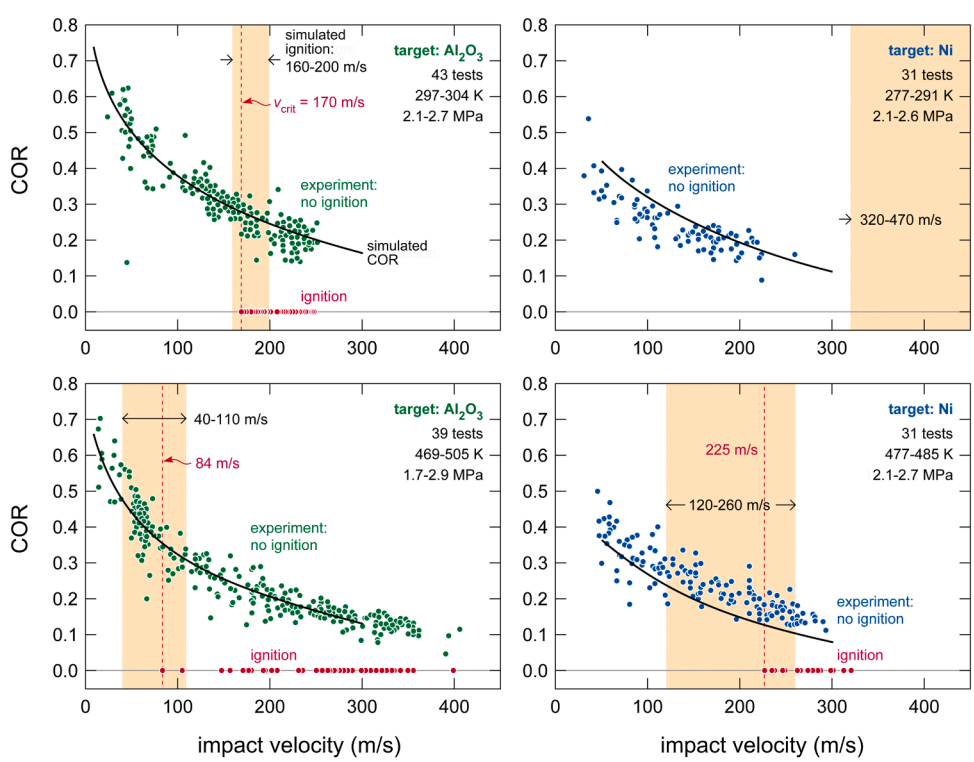

Fig. 5. Particle ignition data and critical velocities for 100-μm Ti64 particles impacting Al_2O_3 and Ni targets at low- and high-temperature conditions, showing that critical velocity is higher on the Ni target and decreases with increasing temperature.

the target produces strong pressure and temperature gradients that vary beyond the nominal estimates, as discussed previously. There are several additional sources of variability, such as particle-to-particle collisions that contribute to particle heating. Altogether, these effects suggest that v_{crit} represents impacts at the most severe conditions and can provide a practical, conservative bound in the context of hardware design.

3. Finite element modeling of particle ignition

Next, we used the experimental results to develop a finite element model (FEM) that describes Ti64 particle ignition upon impact. The FEM assumes a simplified physical picture, as shown in Fig. 6: impact causes rapid particle deformation, adiabatic heating, and rupture of the native surface oxide, which accelerates oxidation and causes thermal runaway. The simulations were carried out using ABAQUS/Explicit with an axisymmetric dynamic explicit model. The particle and target were meshed using 4-node, bilinear axisymmetric quadrilateral elements with reduced integration and hourglass control. The element sizes in the particle and target were 1% of the particle diameter, and interfaces were modeled with general contact using a penalty friction formulation with a friction coefficient of 0.3. An encastre boundary condition was applied to the bottom and outer edges of the target, and we assigned equal initial temperatures to the particle and target.

To model plastic deformation of metal particles and targets, we implemented a Johnson-Cook material model [36], which captures the dependence of equivalent flow stress on strain, strain rate, and temperature. The expression for equivalent flow stress σ_{eq} is given by

$$\sigma_{eq} = [A + B(\varepsilon^p)^n] \times [1 + Cln(\dot{\varepsilon}^p/\dot{\varepsilon}_0)] \times [1 - ((T - T_0)/(T_m - T_0))^m], \quad (2)$$

where ε^p is the equivalent plastic strain, $\dot{\varepsilon}^p$ is the equivalent plastic strain rate, T is temperature, A is the static yield strength, B is the strain hardening modulus, C is the strain rate sensitivity coefficient, n is the strain hardening exponent, m is the thermal softening exponent, T_m is melting temperature, and $\dot{\varepsilon}_0$ and T_0 are reference strain rate and temperature. We assumed that the Al_2O_3 targets remained in the elastic regime during impact due to their high hardness. The simulations considered the material combinations used in experiments, with Ti64 particles impacting Al_2O_3 and Ni targets, and used the material properties given in Table 2. Upon impact, plastic work was converted to adiabatic heating assuming a Taylor-Quinney coefficient of 0.9, in line with typical values for adiabatic heating of metals [37–39]. We also assumed that no heat was transferred between the particle and target during impact given brief contact times around 200 ns.

We evaluated whether the particle ignites by incorporating oxidative and dissipative heat fluxes in the FEM. We expect that the particles have a native oxide consisting primarily of TiO_2 with thickness of just several nanometers, which is easily ruptured during impact [45]. After oxide rupture, oxidation is limited by the rate at which oxygen molecules adsorb on the bare metal surface, estimated using the Hertz-Knudsen expression [46],

$$(1/A)(dm/dt) = (\alpha_s P M^{1/2})/(2\pi R T_p)^{1/2}, \quad (3)$$

where $(1/A)$ (dm/dt) represents the mass flux of oxygen gas adsorbing on the particle surface; α_s represents the sticking coefficient, assumed to be 1; P represents oxygen gas pressure; and M represents the molar mass of O_2.

The resulting heat flux due to oxidation of a bare Ti64 surface, $\dot{q}_{ox}$, is

$$\dot{q}_{ox} = Q(\alpha_s P M^{1/2})/(2\pi R T_p)^{1/2} \times \exp(-E_{A,ad}/(RT_p)), \quad (4)$$

where Q is the enthalpy of oxidation per mass of O_2 consumed, taken to be 29 MJ/kg for Ti64 [47], and $E_{A,ad}$ is the activation energy for O_2 adsorption on Ti64.

For the particle to ignite, this oxidation-driven heat flux must exceed the combined rates of heat loss from conduction into the particle, convection of heat into the surrounding gas, and radiation into the surrounding medium. The largest heat loss initially derives from heat conduction into the particle, where localized deformation drives a steep thermal gradient. This heat flux is given by

$$\dot{q}_{cond} = -k\nabla T_p, \quad (5)$$

where k is the particle thermal conductivity. Heat loss due to convection depends on the temperature difference between the particle and the surrounding gas through

$$\dot{q}_{conv} = h(T_p - T_g), \quad (6)$$

where T_g is the gas temperature and h is the convective heat transfer coefficient, given by $h = (\mathrm{Nu}\, k_{ox}) / D$, with Nu the Nusselt number, k_{ox} the thermal conductivity of O_2, and D the particle diameter. For gas flowing around a sphere, Nu is given by

$$\mathrm{Nu} = 2 + (0.4\mathrm{Re}^{1/2} + 0.06\mathrm{Re}^{2/3})\mathrm{Pr}^{0.4}(\mu/\mu_s)^{1/4}, \quad (7)$$

with Reynolds number $Re = (\rho_g u D)/\mu$, Prandtl number $\mathrm{Pr} = C_{p,g}\mu/k_{ox}$, gas density ρ_g, gas velocity u, dynamic gas viscosity μ, gas specific heat $C_{p,g}$, and reference viscosity μ_s. Here, we use Sutherland's law for O_2 viscosity [48],

$$\mu = \mu_s(T_g/T_{ref})^{3/2}(T_{ref} + S)/(T_g + S), \quad (8)$$

where T_{ref} is a reference temperature and S is the Sutherland temperature. Additionally, we assume that O_2 density ρ_g scales as an ideal gas with the relationship $\rho_g = P\,M / (R\,T_g)$. This approach gives heat transfer coefficients in the range 1.0–2.5 × 10^4 W/m^2·K. Finally, heat loss due to radiation is given by

Table 2
Material parameters used in analysis [40–44].

	Ti64	Al_2O_3	Ni
Density, ρ [kg/m^3]	4430	3980	8900
Young's modulus, E [GPa]	113	416	204.5
Poisson's ratio, ν	0.332	0.231	0.31
A [MPa]	920	-	275
B [MPa]	380	-	3377
C	0.042	-	0.03
n	0.578	-	0.5
m	0.633	-	0.3
reference strain rate, $\dot{\varepsilon}_0$ [s^{-1}]	1	-	1
melting temperature, T_m [K]	1833	-	1728
specific heat, C_p [J/kg·K]	559	755	432.4
thermal conductivity, k [W/m·K]	7	10	66.5

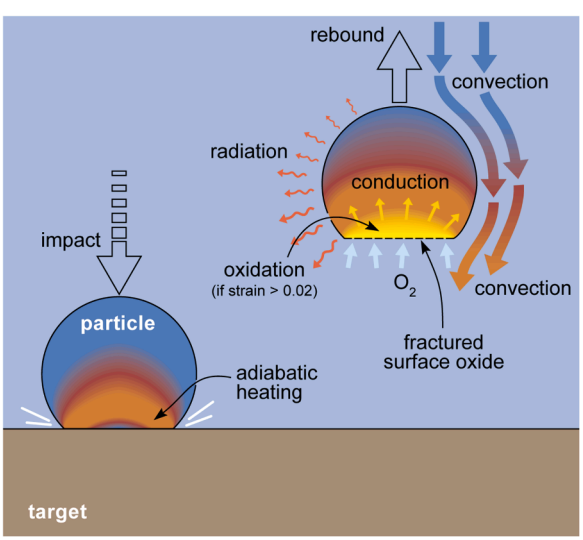

Fig. 6. Finite element modeling captures particle impact, adiabatic heating, and oxide rupture, which cause surface heat fluxes due to oxidation, convection, and radiation and can lead to thermal runaway and ignition.

$$\dot{q}_{\mathrm{rad}} = \varepsilon\sigma\left(T_{\mathrm{p}}{}^{4} - T_{\mathrm{g}}{}^{4}\right), \tag{9}$$

where ε represents particle emissivity, taken as 0.5 as a practical assumption given the range of reported values for metals [49–51], and σ represents the Stefan–Boltzmann constant, equal to 5.67×10^{-8} $W/m^2{\cdot}K^4$. O_2 gas properties used in this analysis are summarized in Table 3.

We incorporated $\dot{q}_{\mathrm{ox}}$, $\dot{q}_{\mathrm{conv}}$, and $\dot{q}_{\mathrm{rad}}$ into the FEM as heat fluxes on the particle surface. Eq. (4) predicts that under relevant temperature and pressure conditions, <1% of the particle mass is consumed over the ignition timescale of tens of μs. Accordingly, we neglect particle mass consumption in the FEM. We assumed an oxide strain to failure of 2% [53], where above this strain the oxide is ruptured and the surface heat flux follows the adsorption-limited oxidation given in Eq. (4). Elsewhere, the oxide is assumed to be intact, and no heat flux is applied. Additionally, we assumed that oxygen was not depleted from the surroundings given the high pressure.

We first validated the material models given in Table 2 by comparing simulated COR measurements to those from experiments, with good agreement as shown in Fig. 5. Having validated the material models, we implemented the heat fluxes described earlier to simulate particle impact and observe the resulting temperature evolution. Exemplary temperature trajectories are shown in Fig. 7, with initial temperature 300 K, O_2 pressure 2.5 MPa, and an Al_2O_3 target. With impact velocity 150 m/s, the particle does not ignite: the particle is adiabatically heated to 510 K at the impact site, forming a steep thermal gradient that quickly dissipates via thermal conduction. The peak temperature reduces to 450 K within 2.5 μs and then continues to decrease as convective cooling overtakes oxidative heating. By contrast, the particle ignites with impact velocity 200 m/s: the particle reaches a peak temperature of 589 K upon impact, and heat is quickly conducted from the impact site as the temperature reduces to 525 K within 1.5 μs. Oxidation gradually overtakes convection as the peak temperature slowly rises, and thermal runaway sets on around 20 μs as oxidation accelerates. The particle reaches its melting temperature around 25 μs, where we assume that the particle has ignited and terminate the simulation. These simulated timescales align with experiments (cf. Fig. 4), where high-speed videos show that particle ignition occurs within one frame, or 34 μs. In Fig. 7, thermal runaway appears to set on when the peak temperature reaches around 600 K, below the reported temperature for static Ti64 particle ignition between 800 and 900 K [54]. This result aligns with previous studies that show that mechanical processes can cause ignition at relatively low temperatures [30].

We calibrated the activation energy $E_{\mathrm{A,ad}}$ in Eq. (4) by predicting critical velocity ranges and comparing them to the experimentally determined critical velocities. We performed these simulations for the high- and low-temperature test conditions under the ranges of temperatures and pressures given in Table 1, considering the full range of particle sizes from 90 to 106 μm. For each bounding combination of temperature, pressure, and particle size, we found a critical velocity using a binary search algorithm with an accuracy of 10 m/s, which gave an upper and lower bound on critical velocity for each test condition and target material. Using $E_{\mathrm{A,ad}} = 30$ kJ/mol gives good agreement between the predicted critical velocities and those measured in experiments, as shown in Fig. 5. This calibrated $E_{\mathrm{A,ad}}$ is slightly less than the reported range of 34–55 kJ/mol for mechanical ignition of Ti64 via fracture [52], which reflects that the critical velocity measurements are intrinsically conservative in that they represent the lower bound on ignition velocities.

Table 3
O_2 gas properties used in analysis [52].

Property [units]	Value
sticking coefficient, α	1
molar mass, M [kg/mol]	0.032
thermal conductivity, k_{ox} [W/m·K]	0.046
gas velocity, u [m/s]	50
O_2 specific heat, $C_{\mathrm{p,g}}$ [J/kg·K]	918
reference viscosity, μ_{s} [Pa·s]	2.07×10^{-5}
reference temperature, T_{ref} [K]	273
Sutherland temperature, S [K]	127

The predictions agree well with experiments and capture the key trends observed previously; i.e., critical velocity is lower for the Al_2O_3 target and at the high-temperature condition. Additionally, Fig. 5 shows that the FEM predicts wider critical velocity ranges for softer targets, where much of the impact energy is dissipated through deformation and heating of the target. The agreement between the experiments and predictions indicates that the FEM captures the physical mechanisms causing particle ignition: deformation ruptures the native oxide and causes local adiabatic heating, leading to rapid oxidation; the resulting competition between heat fluxes—dominated by oxidation—leads to thermal runaway.

4. Extrapolation of particle ignition model to engine conditions

We extended the FEM to predict critical velocity over ranges of temperature (60–750 K), pressure (0.1–69 MPa), and particle size (50–500 μm) relevant to the rocket engine turbopump application. Select results are summarized in Fig. 8, which shows critical impact velocities for Ti64 particles impacting Al_2O_3 as a function of particle size and temperature under an oxygen pressure of 2.5 MPa, with experimental data included for comparison. Fig. 9 extends the analysis to an oxygen pressure of 69 MPa, representative of conditions encountered in rocket engines and currently inaccessible through experiments. The analysis predicts that critical velocity decreases as temperature, pressure, and particle size increase. As temperature increases, particles thermally soften and undergo more deformation and heating upon impact. Oxidation also accelerates as temperature increases according to the Arrhenius-type relationship given in Eq. (4). Increasing pressure results in competing effects: the rate of convective heat transfer increases according to Eq. (6), but it is predicted that this effect is outweighed by the increase in oxygen adsorption and oxidative heating. Critical velocity also decreases with particle size. Large particles experience a lower strain rate at a given impact velocity, as strain rate scales inversely with particle diameter [55]. The reduced strain-rate hardening increases the total plastic strain and thus the total heat generated during impact [56].

Figs. 8 and 9 also show that the critical velocities tend toward low values at elevated temperatures. With Al_2O_3 targets, the critical velocity becomes negligible above approximately 600 K and 300 K at 2.5 MPa and 69 MPa, respectively. Under these conditions, particle ignition is predicted to occur as long as the native oxide layer is ruptured, enabling the onset of rapid oxidation. By comparison, with Ni targets at 69 MPa, the critical velocity approaches roughly 100 m/s above 400 K. Here, the particles require a higher impact velocity on the softer Ni target to rupture their native oxide and cause thermal runaway. Similar results were demonstrated in previous work which showed that soft Ni and glass-ceramic targets increase the critical velocity for Ti64 particle ignition as compared to Al_2O_3 targets [57], motivating the use of soft coatings to mitigate particle ignition in service.

These predictions explain qualitative trends that were identified in previous particle impact ignition studies. For example, several studies observed through experiments that the likelihood of ignition increases with particle size, considering Al particles, various target materials, and particle sizes from 300 to 2000 μm [4,7,18,19]. Previous tests also showed that higher gas pressures can depress the ignition temperature [4]. The FEM directly captures this effect by incorporating the pressure-dependent heat flux given in Eq. (4).

Over the full range of parameters, the model highlights significant particle ignition risk above around 300 K and 6 MPa, where predicted critical velocities for impact on the hard Al_2O_3 target are roughly 20–60 m/s for particles larger than 100 μm. For comparison, gas velocities in

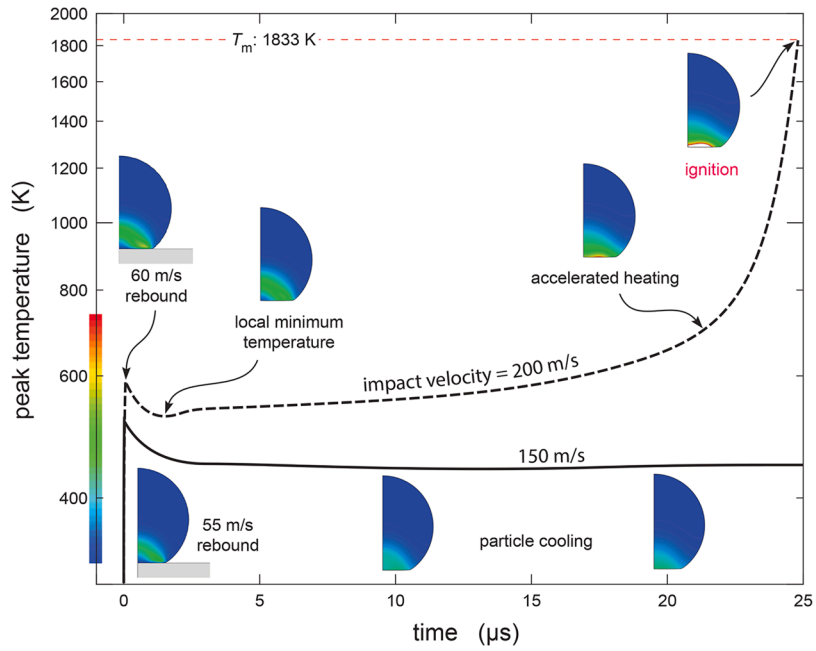

Fig. 7. Simulated 100-μm Ti64 particle impact on Al_2O_3 with initial temperature 300 K showing no ignition with impact velocity 150 m/s, and ignition with impact velocity 200 m/s.

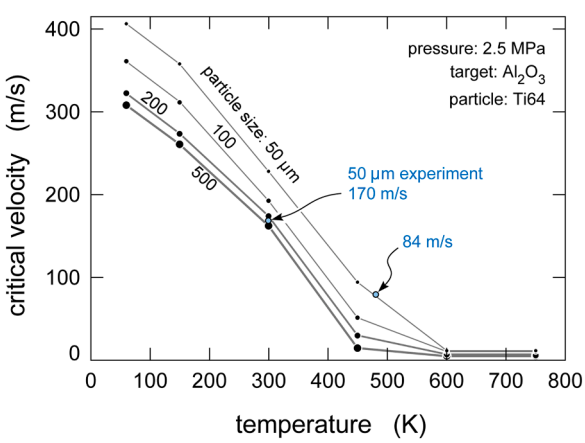

Fig. 8. Critical velocity predictions for Ti64 impact on Al_2O_3 for a range of particle sizes and initial temperatures at 2.5 MPa.

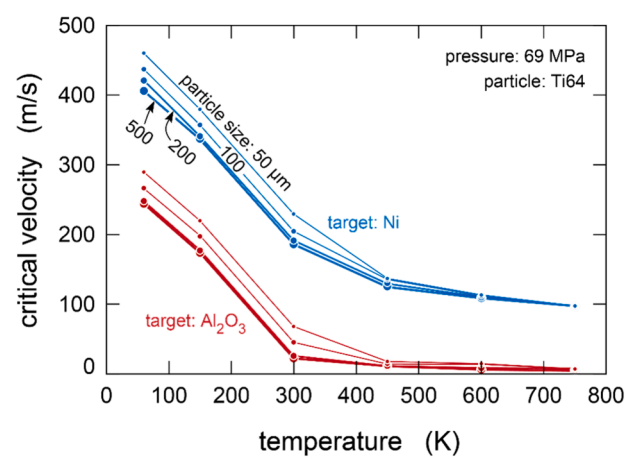

Fig. 9. Critical velocity predictions for Ti64 impact on Al_2O_3 and Ni for a range of particle sizes and initial temperatures at 69 MPa, which corresponds to rocket engine operating conditions.

the turbine of rocket engine turbopumps can easily exceed 100 m/s, suggesting that particles in these environments will readily ignite. This risk is compounded by particle-size effects, where increasing particle size from 50 μm to 200 μm can reduce the critical velocity by up to 40%. Accordingly, filtering and cleanliness standards need to consider the trade-off between ignition risk and engine performance with increasing allowable particle size. These results also motivate the use of soft coatings (e.g., pure Ni [58]) to mitigate particle ignition in locations with slow gas velocities. However, our results indicate that at higher gas velocities (above ~200 m/s), ignition can occur even with a soft coating present. Accordingly, these conditions call instead for inert coatings, such as the oxide glass-ceramic coating discussed in [57], to prevent kindling of a burning particle to engine hardware.

5. Conclusions

The key conclusions of this work are as follows:

- Particle impact tests in flowing O_2 reveal critical impact velocities for Ti64 particles on Ni and Al_2O_3 targets. At ~300 K, no particle ignition events are observed on Ni, while the critical velocity on Al_2O_3 is 170 m/s; at ~500 K, the respective critical velocities are 225 m/s and 84 m/s. Overall, critical velocity decreases with increasing temperature and target hardness, which demonstrates coupling between thermal softening, oxidation, and ignition.
- Numerical modeling incorporates plastic heating, oxide rupture, oxygen adsorption, and convective/conductive cooling, and demonstrates a transition to thermal runaway above a critical impact velocity. The predicted critical velocities, ignition delay times, and localized ignition modes align with experiments, confirming that the FEM captures key physics linking particle impact to ignition.
- Extending the FEM to temperatures, pressures, and particle sizes relevant to operation predicts high ignition risk above 600 K and 6 MPa, which are common conditions in oxygen-rich turbomachinery. In these conditions, critical velocities for hard targets are only 20–60

m/s, well below the gas velocities expected in rocket turbines (>100 m/s). This finding highlights that small particulate and wear debris present significant ignition risk in operation.

- The combined data and model directly inform risk mitigation techniques. Filtering and cleanliness standards should consider the trade-off between ignition risk and engine performance with increasing allowable particle size. The results also highlight the benefit and potential limitations of using soft coatings to mitigate particle ignition in operation. While soft Ni coatings may offer protection at low gas velocities (below 200 m/s), they do not prevent particle ignition at high gas velocities. These conditions instead require inert coatings to prevent kindling of a burning particle to engine components.

The experimentally calibrated FEM in this work presents the first quantitative link between impact and particle ignition conditions. While this work focuses on particle ignition, ongoing work integrates the FEM into a component-level risk analysis framework that predicts the likelihood of particle impact ignition in complex geometries, accelerating the design and qualification process for high-pressure oxygen-rich systems.

CRediT authorship contribution statement

Spencer V. Taylor: Writing – original draft, Investigation, Formal analysis, Conceptualization. **Bodie J. Ziertman:** Investigation. **Carmine S. Taglienti:** Investigation. **Steven A. Mathe:** Methodology, Investigation. **Jonathan M. Tylka:** Methodology, Investigation. **Stephen F. Peralta:** Methodology, Investigation. **Gregory J. Harrigan:** Supervision, Project administration. **Zachary C. Cordero:** Writing – review & editing, Supervision, Project administration, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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